

CLOSING THE PERMANENT-RANK BOUNDARY FOR THE ALON-JAEGER-TARSI CONJECTURE

ZIJIAN ZENG, HOUDE LIU, AND KURU RATNAVELU

ABSTRACT. Let $p \geq 5$ be prime. The theorem of Yu obtained from permanent rank proves the Alon–Jaeger–Tarsi conjecture over $\mathbb{F}_p$ in dimensions $n < 2^{p-2}$. We close the missing boundary: the conjecture holds for every $n \leq 2^{p-2}$. In particular, for every $M \in \text{GL}_8(\mathbb{F}_5)$ there is a vector $x \in (\mathbb{F}_5^*)^8$ such that $Mx \in (\mathbb{F}_5^*)^8$. The proof separates according to the permanent rank of M . Above half rank, Yu’s concatenation recurrence and the Combinatorial Nullstellensatz force a nowhere-zero point. At equality, the rigidity theorem of Kisley and Shader reduces M , under monomial equivalence, to a direct sum of 2×2 blocks, for which a nowhere-zero point is explicit.

1. INTRODUCTION

The Alon–Jaeger–Tarsi conjecture asks whether, for every finite field $\mathbb{F}$ with $|\mathbb{F}| \geq 4$ and every $M \in \text{GL}_n(\mathbb{F})$, there is a vector $x \in (\mathbb{F}^*)^n$ such that $Mx \in (\mathbb{F}^*)^n$; see [1]. The non-prime finite fields were settled in the original work, and Nagy and Pach recently proved the conjecture for primes $p > 61$, $p \neq 79$ [4]. The small prime fields remain the main unresolved range.

For a matrix A , write $\text{prank } A$ for the largest order of a square submatrix of A having nonzero permanent. Yu proved

$$(1) \quad \text{prank } A \geq \frac{\text{rank } A}{2}$$

and used a concatenation argument to establish the Alon–Jaeger–Tarsi conjecture over $\mathbb{F}_p$ whenever $n < 2^{p-2}$ [5]. For $p = 5$ this leaves $n = 8$ as the first dimension outside that theorem. A 2026 sequel of Yu still presents the $\mathbb{F}_5$ assertion as a conjecture while noting the recent classification of equality in (1) [6].

Our observation is that the strict inequality in Yu’s dimension bound can be removed. If $\text{prank } M$ is strictly larger than half the rank, the integer rounding in Yu’s recurrence reaches full permanent rank at the boundary. If equality holds, a rigidity theorem of Kisley and Shader identifies the matrix up to transformations that preserve the algebraic torus [3, Theorem 4.8].

2020 *Mathematics Subject Classification*. Primary 15A15; Secondary 05A05, 11T06.

Key words and phrases. Alon–Jaeger–Tarsi conjecture, permanent rank, Combinatorial Nullstellensatz, nowhere-zero point, finite field.

Theorem 1.1. *Let $p \geq 5$ be prime and $n \leq 2^{p-2}$. For every $M \in \text{GL}_n(\mathbb{F}_p)$ there exists $x \in (\mathbb{F}_p^*)^n$ such that $Mx \in (\mathbb{F}_p^*)^n$.*

Corollary 1.2. *For every $M \in \text{GL}_8(\mathbb{F}_5)$ there exists $x \in (\mathbb{F}_5^*)^8$ such that $Mx \in (\mathbb{F}_5^*)^8$.*

The proof is entirely algebraic. We include the short concatenation and coefficient arguments to make clear exactly where the boundary is gained.

2. A PERMANENT-RANK RECURRENCE

For matrices having the same number of rows, $[B \ A]$ denotes horizontal concatenation. We first isolate the one-step estimate implicit in Yu's proof.

Lemma 2.1 (Yu's recurrence). *Let B be an $n \times N$ matrix over a field of characteristic different from 2, and let $A \in \text{GL}_n(\mathbb{F})$. If $r = \text{prank } B$, then*

$$\text{prank}[B \ A] \geq \left\lceil \frac{r+n}{2} \right\rceil.$$

Proof. The statement is immediate if $r = n$. Otherwise select r columns of B forming an $n \times r$ matrix D_0 with full permanent rank. Enlarge D_0 by a maximal subset of the columns of A such that the resulting $n \times s$ matrix D has permanent rank s .

If $s = n$, the conclusion is immediate, so assume $s < n$.

Yu's column-appending lemma [5, Lemma 1.1] states that

$$L(D) = \{z \in \mathbb{F}^n : \text{prank}[D \ z] = s\}$$

is a linear subspace of dimension at most s . By maximality, every column of A not selected for D belongs to $L(D)$. There are $s - r$ selected columns of A , and all the columns of A span $\mathbb{F}^n$. Therefore

$$n \leq (s - r) + \dim L(D) \leq 2s - r.$$

Thus $s \geq (r + n)/2$, and integrality gives the result. $\square$

For $A \in \text{GL}_n(\mathbb{F})$, let $A^{[j]} = [A \ \cdots \ A]$ be the concatenation of j copies of A , and set

$$k_j = \text{prank } A^{[j]}, \quad d_j = n - k_j.$$

Then Theorem 2.1 gives the useful deficiency recurrence

$$(2) \quad d_{j+1} \leq \left\lfloor \frac{d_j}{2} \right\rfloor.$$

3. FROM FULL PERMANENT RANK TO A TORUS POINT

The next lemma is the standard polynomial implication behind Yu's argument. We give the coefficient calculation because the restriction to $p - 2$ copies is essential.

Lemma 3.1. *Let $A \in M_n(\mathbb{F}_p)$, where p is prime. If $\text{prank } A^{[p-2]} = n$, then there exists $x \in (\mathbb{F}_p^*)^n$ such that $Ax \in (\mathbb{F}_p^*)^n$.*

Proof. Write $A = (a_{ij})$ and consider the homogeneous polynomial

$$g(X_1, \dots, X_n) = \prod_{i=1}^n \left(\sum_{j=1}^n a_{ij} X_j \right).$$

There is an $n \times n$ submatrix N of $A^{[p-2]}$ with $\text{per } N \neq 0$. Let e_j be the multiplicity with which the j th column of A occurs in N . Then

$$0 \leq e_j \leq p-2, \quad \sum_{j=1}^n e_j = n.$$

Labeling the repeated columns in the permanent expansion gives

$$(3) \quad \text{per } N = \left(\prod_{j=1}^n e_j! \right) [X_1^{e_1} \cdots X_n^{e_n}] g.$$

The factorials are nonzero in $\mathbb{F}_p$, so the coefficient on the right-hand side is nonzero. Apply the Combinatorial Nullstellensatz [2] to the sets $S_j = \mathbb{F}_p^*$. Since $|S_j| = p-1 > e_j$, there exists $x \in \prod_j S_j$ with $g(x) \neq 0$. Equivalently, all coordinates of both x and Ax are nonzero. $\square$

4. THE EQUALITY CASE AND THE BOUNDARY

Let

$$P_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

Two square matrices are *monomially equivalent* if one is obtained from the other by multiplying on the left and right by invertible monomial matrices. Such matrices permute and rescale coordinates, hence biject the algebraic torus $(\mathbb{F}^*)^n$ with itself and preserve the nowhere-zero property.

Kisley and Shader proved that, over a field of characteristic different from 2, every invertible matrix A satisfying $\text{prank } A = \text{rank}(A)/2$ is monomially equivalent to a direct sum of copies of P_2 [3, Theorem 4.8]. For $p \geq 5$ the vector $(1, 2)^T$ is a nowhere-zero input for P_2 , because

$$P_2 \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} \in (\mathbb{F}_p^*)^2.$$

Consequently every invertible equality-case matrix has a nowhere-zero point.

Proof of Theorem 1.1. Let $k_1 = \text{prank } M$. By (1), $k_1 \geq n/2$.

If $k_1 = n/2$, the Kisley–Shader classification and the block construction above give the desired point.

Suppose $k_1 > n/2$. Then the integer $d_1 = n - k_1$ satisfies

$$d_1 < \frac{n}{2} \leq 2^{p-3},$$

and hence $d_1 \leq 2^{p-3} - 1$. Iterating (2) through the remaining $p - 3$ copies gives

$$d_{p-2} \leq \left\lfloor \frac{d_1}{2^{p-3}} \right\rfloor = 0.$$

Thus $\text{prank } M^{[p-2]} = n$, and Theorem 3.1 finishes the proof. $\square$

Proof of Theorem 1.2. Here $p = 5$ and $n = 8 = 2^{p-2}$. More explicitly, if $\text{prank } M = 4$, the equality classification applies. If $\text{prank } M \geq 5$, then Theorem 2.1 yields

$$\text{prank}[M \ M] \geq \left\lceil \frac{5+8}{2} \right\rceil = 7, \quad \text{prank}[M \ M \ M] \geq \left\lceil \frac{7+8}{2} \right\rceil = 8,$$

so Theorem 3.1 applies. $\square$

Remark 4.1. The hypothesis $p \geq 5$ is used in the equality case. Over $\mathbb{F}_3$, the two ratios ± 1 exhaust $\mathbb{F}_3^*$, and P_2 itself has no nowhere-zero input.

DATA AND CODE AVAILABILITY

No computation is used in the proof. An exact-arithmetic companion verifier checks the exceptional 8×8 normal form, its permanent rank and explicit witness, as well as all repeated-column coefficient identities occurring for that normal form. The verifier and its machine-readable output are included with the source package.

REFERENCES

- [1] N. Alon and M. Tarsi. A nowhere-zero point in linear mappings. *Combinatorica*, 9(4):393–395, December 1989.
- [2] NOGA ALON. Combinatorial nullstellensatz. *Combinatorics, Probability and Computing*, 8(1-2):7–29, January 1999.
- [3] Ben Kisley and Bryan Shader. Matrices whose permanent rank equals half their rank. *Communications in Optimization Theory*, 2025:49, 2025. Article 49, 14 pages.
- [4] János Nagy and Péter Pál Pach. The alon–jaeger–tarsi conjecture via group ring identities. *Journal of the European Mathematical Society*, April 2025.
- [5] Yang Yu. The permanent rank of a matrix. *Journal of Combinatorial Theory, Series A*, 85(2):237–242, February 1999.
- [6] Yang Yu. The permanent rank of a matrix (part two). Preprints.org manuscript 202603.0035.v1, March 2026. Not peer reviewed.

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR 56000, MALAYSIA

Email address: 1002266693@ucsiuniversity.edu.my

TSINGHUA SHENZHEN INTERNATIONAL GRADUATE SCHOOL, SHENZHEN 518055, CHINA

Email address: liu.hd@sz.tsinghua.edu.cn

INSTITUTE OF COMPUTER SCIENCE AND DIGITAL INNOVATION, UCSI UNIVERSITY,
KUALA LUMPUR 56000, MALAYSIA

Current address: Institute of Mathematical Sciences, University of Malaya, Kuala Lumpur 50603, Malaysia

Email address: Kurunathan@ucsiuniversity.edu.my